

Blockchain based federated learning model for Pneumonia detection using Chest X-rays

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Abstract—Federated Learning (FL) has emerged as an efficient technology in the field of healthcare. Blockchain has been used to address the issues that the traditional FL has. This paper examines the incorporation of Blockchain into federated learning systems and its advantages by building a model for the same. Different aggregation techniques (FedAvg, FedBoosting, and Geometric Mean) are explored and implemented in a decentralized manner using a public blockchain network. The accuracy of the local models and the global model is studied and compared. The global model's accuracy is seen to be higher. Future developments have been discussed.

Index Terms—Blockchain-based Federated Learning, Pneumonia, Smart Contract

I. INTRODUCTION

The usage of neural network models has proven to have incredible promise for spotting complex linkages and patterns in images. The performance of these models can be greatly improved by creating deeper models. However, a sizable training dataset is required to build a trustworthy and efficient deep learning model that can function well in a variety of circumstances. Traditionally, data is gathered and processed in one location. This necessitates the transmission of data to the main training data centre from numerous places. It takes a lot of resources to gather enough data and train a deep learning model, and there are issues with how to store it all safely.

The growing amount of personal data being gathered by different applications has heightened the need for establishing appropriate policies and rules regarding where data storage and processing can take place. Many countries have also implemented regulations to protect individual privacy. As a result, the location of data centers is now restricted, which limits the ability to collect data in a centralized location for processing.

Federated Learning (FL) can be useful in this situation. Without centralised training data, federated learning is a co-operative machine learning approach. [9] Federated Learning operates differently from traditional machine learning, which brings code to the data. This method uses several decentralised nodes with local data samples to train an algorithm without exchanging the samples. Unlike the conventional ML framework, FL only needs model parameters to be uploaded; clients, or the owners of the data, are not required to send the raw data to the central server for model training. The server receives these

local model parameters, which are then combined to produce the global model. The local models are then updated after the global parameters have been communicated back to them. The updated local models can be used for accurate prediction on the client side.

FL still has its shortcomings [11] :

- **Single Point of Failure:** In the Federated Learning (FL) method, the outcomes of local training are combined and updated on a central server known as an aggregator. The aggregator is not always reliable, though. The FL system as a whole might malfunction if the aggregator is compromised. The FL system may experience challenges with the aggregator due to things like willful dishonest aggregation, unintentional network disconnections, and unanticipated external attacks.
- **Malicious clients and false data :** Due to the large number of participants in FL, it is not reasonable to assume that every client will be reliable and adhere to FL guidelines while training their local models. False information provided by some clients regarding their training outcomes may have an impact on the performance of the overall model. Malicious clients who train their own models with incomplete datasets may also target the FL system. The performance of the global model can be significantly impacted by the existence of erroneous data.
- **The lack of incentives/Rewards:** The conventional FL approach assumes that clients will contribute their computing power without receiving any compensation, which makes it challenging to motivate them to participate honestly and provide trustworthy data. Additionally, FL often requires multiple devices to work together, particularly for data-intensive training tasks that need a significant number of participants. However, the traditional FL framework may not attract enough clients to engage in FL training because of the absence of incentives.

To address these issues, we present a novel approach that leverages blockchain technology to create a secure and trustworthy environment for federated learning. We outline the proposed model's architecture, which consists of a federated learning layer and a blockchain layer (which incorporates a smart contract). Using a real-world dataset, we assess the

model's performance in terms of data privacy, security, and trust. A promising solution for safe and private machine learning on decentralised data is offered by our proposed blockchain-based federated learning model.

In this project, we are using a neural network to train the local model at each hospital with the hospital's local data. Image recognition is used to classify Chest X-rays. After the local model is built, the model parameters are sent to the smart contract inside the blockchain, which makes sure that the aggregation is happening in a decentralized manner. After the global model is built, the global model parameters are stored in the blockchain which can be accessed by the hospitals when required. The global model can be used by each Hospital client to predict and analyze any patient's chest X-ray.

The process of automating model training and aggregation can be done with **smart contracts**. A smart contract, for instance, can describe the training algorithm to be used, its parameters, the number of repetitions, and the aggregation technique. By doing this, the training procedure can be made transparent and standardised. In BCFL, a smart contract's main function is to perform decentralised aggregation on dispersed data such that the need for a central server is no longer necessary. After obtaining the local model parameters and reaching consensus, the smart contract can apply any one of the aggregation methods that are accessible.

II. LITERATURE SURVEY

Federated Learning was first introduced by Google as a distributed machine learning paradigm to train the model with data available locally from devices while ensuring privacy as seen in the paper by McMahan [1]. The data used for model training are broadcasted from various participating companies and users to converge the FL model. Multiple papers like [8] are being published to discuss the remarkable use of FL in the real world, particularly in healthcare, finance, transportation, and smart cities, to mention a few.

Toyoda [2] explains the need to incorporate blockchain in federated learning systems. He details that even though FL outperforms the performance of traditional ML, it has a high risk of SPoF and DDoS attacks. Consequently, a powerful decentralised system is needed to identify and stop rogue updates. Blockchain offers a tremendous deal of potential to draw model training participants by including a digital currency.

In literature, several studies are available related to federated learning (2020), and blockchain-based federated learning approaches (2021) discussed the blockchain-based federated learning (BCFL) architecture with respect to types, design, model improvement, and incentive mechanisms. In 2022, papers [13] are exploring different blockchain deployment frameworks and dependencies and aggregation algorithms.

In the healthcare area, the data of patients are sensitive thus both patients and hospitals are reluctant to share their health data. FL can help train the model distributively, while data leakage is the biggest challenge [7]. Blockchain can be implemented among patients or hospitals, allowing participants to share data without privacy disclosure. Passerat-Palmbach et al. [4] point out that the protection of patient privacy constrains researchers from analyzing health data, and the existing tools are insufficient to address the issue, so they suggest to use both blockchain and FL for healthcare consortia. In their model, data access, model integration, weight encryption, and auditing of the learning process are emphasized. However, this study is specific to consortia and is not appropriate for most health problems and lacks concrete solutions which can be operated. In contrast to [4], Kumar et al. [5] offer a specific solution for COVID-19 detection via BCFL models. Hospitals train the local model based on their own private data and share only the weights and gradients, and blockchain records the learning process and related data. Researchers highlight the privacy of patients, and BCFL framework can protect privacy when the global model is trained. That paper builds a secure and decentralized data-sharing platform among hospitals, enabling the automatic detection of COVID-19 in a secure manner. Research using BCFL in healthcare are rare now, but the research direction is promising since a large amount of medical data have to proceed and BCFL can offer secure learning environments.

III. ARCHITECTURAL DESIGN FOR THE PROPOSED SYSTEM

A. Data Sources and Federated Learning

In this instance, the data comes from hospitals and is comprised of chest X-ray scans. The many hospitals that take part in federated learning by contributing their chest X-ray pictures for model training are represented by the "Data Sources" component of the architecture diagram for the blockchain-based federated learning system. To guarantee consistency in the image data across many hospitals, the chest X-ray images would first need to be pre-processed and labelled. Once the photos are ready, a federated learning approach would be used to train the machine learning model utilising the hospitals' encrypted chest X-ray images. In our scenario, each participating hospital would create the local parameters and train a local model on its own data at this level. The tasks carried out at this level include data collection, data preprocessing, model training, model update and model evaluation.

B. Blockchain Network

The model aggregation takes place in the smart contract at the blockchain level in our federated learning system built on blockchain technology. A crucial component of providing safe and transparent model updates is the blockchain network. The participating hospitals use their own chest X-ray pictures to train local machine learning models, and instead of sharing the updated model weights with the blockchain network, they

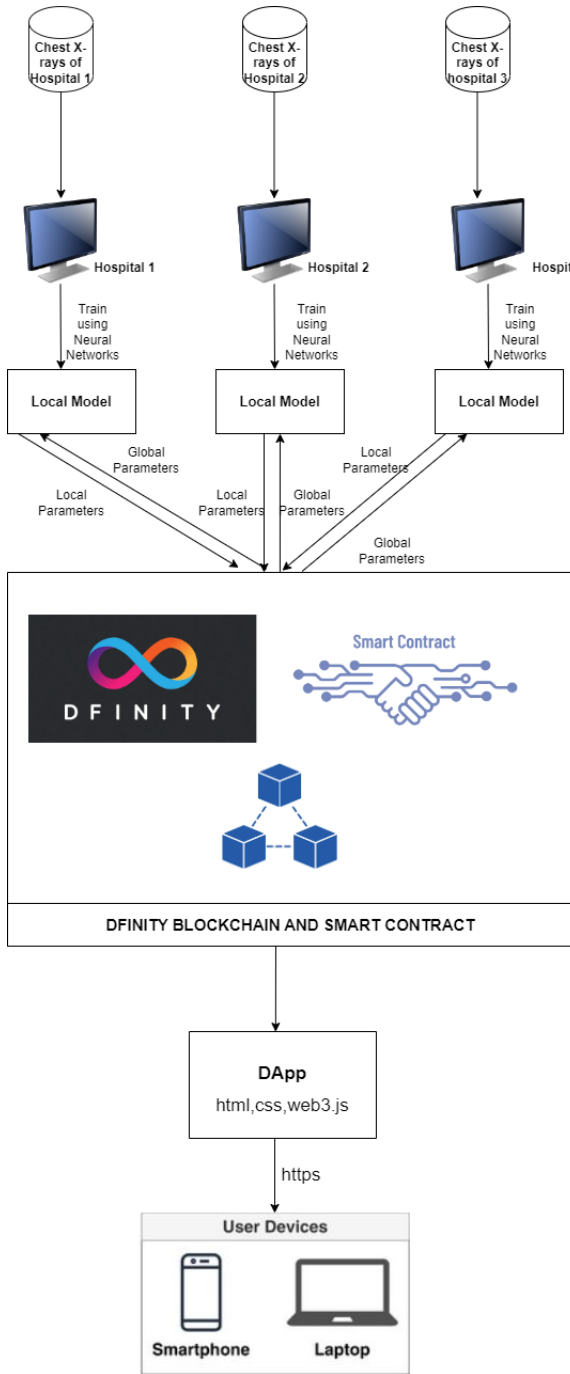


Fig. 1. Architecture Diagram

share the updated model weights. The blockchain network creates a global model by combining the model updates from all involved hospitals.

Here are some key features of the blockchain network component in the architecture:

Decentralization: Since there is no single entity in charge of data control on the blockchain network like a central server, it is decentralised and safe. Instead, each node in the network keeps a copy of the ledger, allowing the hospitals to securely and decentralise the sharing of new model weights.

Consensus Mechanism: DFINITY has its own consensus algorithm. It ensures that all participants in the blockchain network are in agreement with the transactions taking place. This approach makes guarantee that the hospitals' shared data is accurate and that no node can change it without the consent of other nodes.

Smart Contract: A smart contract that is part of the blockchain network specifies the guidelines for taking part in the federated learning process. A model aggregation process is part of the smart contract, therefore the blockchain network is in charge of compiling the updated model weights from all participating hospitals and creating a global model. Different strategies, such as Federated Averaging or Secure aggregate, can be used to execute the aggregate process, allowing the network to produce an accurate global model without jeopardising the privacy of the local data.

Data Sharing: Hospitals can collaborate on model training and share local model parameters in the blockchain network without compromising patient privacy. The hospitals can access the global model that is kept on the ledger, but they are unable to access the real data that other hospitals have provided to train their own local models.

IV. PROPOSED SYSTEM

We propose a system that consists of more than 2 hospital clients with their own Chest X-ray data. Each hospital will train its local model using Convolutional neural network algorithm for image recognition. The local parameters are sent to the hyperledger(blockchain) which aggregates the local parameters without any centralization. The global parameters are sent back to the hospitals which are later used for disease prediction.

The workflow is as follows :

- Clients collect data and train the models locally.
- Local model updates are verified by the (selected) clients.
- Verified local updates are collected by (selected) clients and then the global model will be updated.
- New block which stores the verified model updates is added into the distributed ledger.

A. CNN (Convolutional Neural Network) for Local Model training

A **Convolutional Neural Network (CNN)** is a deep learning algorithm. The key idea behind a CNN is to extract features from an input image using convolutional layers. Python Libraries such Tensorflow, Keras, Conv2D etc... are used to achieve the above.

B. Motoko Smart Contract - Canister

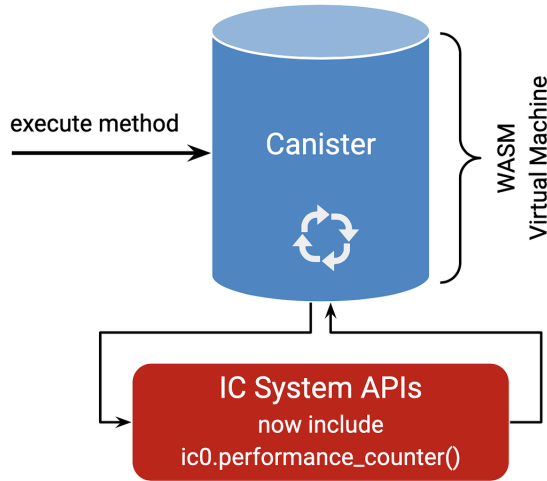


Fig. 2. Architecture Diagram

The **smart contract** in the suggested system is created in Motoko. The smart contract accepts model parameters from new clients and registers them as clients. The collection of local model parameters and the creation of fresh global model parameters are done by the smart contract.

After the code has been developed, it can be uploaded to the Internet Computer blockchain as a canister, a safe container that houses the resources required to run the programme along with the smart contract code and data. When a smart contract is launched, it may be accessed and used by other Internet Computer containers or by outside clients via a predefined interface that enables the clients to access the global parameters from the Internet Computer Blockchain.

C. Aggregation Methods

In this paper, We explored three different aggregation methods and analyzed each one of them.

- **FedAvg**: In FedAvg, the aggregation of the local model's parameters takes place by averaging the parameters and sending the updated model back to the participating devices. This process is repeated multiple times until the model converges to a satisfactory level. FedAvg has been widely adopted in federated learning because it is a simple and effective method to combine local models. FedAvg works well for evenly distributed data.
- **FedBoosting** : In Federated Boosting, the aggregation method used to combine the models is based on the idea of weighted averaging. The weights are determined

by either the performance of each model or the importance/priority attached to each client on a validation set, and the models with better stance are given more weight in the final aggregation.

- **Geometric Mean** : In the Geometric Mean Federated Learning Algorithm, each device trains a local model on its own dataset, and then the geometric mean of the models is calculated to produce a global model. This is done by taking the product of the weights of each model and then taking the n th root of the product, where n is the number of local models. It is less sensitive to outliers or extreme values in the local models, which can lead to more robust global models.

D. CANDID API - The Connection

Candid refers to the public interface of a programme operating as a deployed canister smart contract in the context of the Internet Computer. Candid facilitates communication between front-ends and services created in several programming languages, including JavaScript, Motoko, and Rust. In our system, the canister installed on the internet computer blockchain and the client Python code are in connection thanks to the candid API.

Candid is an Interface Description Language (IDL). It is more desirable over other languages like JSON, XML which are Data Description Languages (DDL) because of the following features:

- Candid doesn't only map individual values to bytes or characters, it also describes the services as a whole.
- Instead of concentrating on the data types, the attention is on the methods that use the data types.
- The types and values of the host language are directly mapped to the Candid value.

V. HEALTHCARE AND SECURITY ISSUES ADDRESSED

Federated learning powered by blockchain has the ability to rectify a variety of hospital system-related healthcare difficulties. The following healthcare conditions can be addressed with the use of this technology

1) **Data Privacy and Security**: Data security and privacy is one of the biggest problems in healthcare. Federated learning makes it possible for medical facilities to work together and exchange parametric data while maintaining the privacy of sensitive data. Without the need to disclose private patient data, blockchain offers a decentralised environment for the compilation of local models created by each client.

2) **Disease Diagnosis and Treatment**: Hospitals may collaborate and share patient data to better diagnose and treat diseases because to federated learning. Doctors might see patterns and learn insights that help them create more successful treatment regimens by combining data from various facilities.

3) **Medical Research**: There is a huge data requirement

for medical research. Hospitals can cooperate and exchange data using federated learning to speed up medical research. Researchers can acquire insights by combining data from many hospitals that would be impossible to get from just one hospital's data.

VI. CONCLUSIONS

The dataset is organized into 3 folders (train, test, val) and contains subfolders for each image category (Pneumonia/Normal). There are 5,863 X-Ray images (JPEG) and 2 categories (Pneumonia/Normal). The dataset is further divided into 3 parts for our three hospital clients.

The system was tested on three systems as three hospitals. Each system was trained on its local data. The model parameter extraction and encoding of the parameters happen at this level. The individual systems then sent the encoded parameters to the canister deployed on 'Motoko Playground'. Motoko Playground is an online platform that provides a web-based interface for developers to write, test, and execute code in the Motoko programming language. It provides 'cycles' of 20 minutes after which the canister can be re-deployed with a new canister id. Hosting it on Internet Computer blockchain increases the security by many folds as it is completely decentralized compared to traditional FL.

The local models were found to have an accuracy of 76%, 82% and 84%. After the parameter aggregation using different aggregation methods and updation of local model parameters with the new global model parameters, the new model accuracy was improved. Below is a table depicting the accuracy analysis for aggregation algorithms used and of the local models.

TABLE I
RESULT ANALYSIS

Models	Accuracy
Local Model 1	76%
Local Model 2	82%
Local Model 3	84%
Global Model (FedAvg)	85%
Global Model (Geometric Mean)	85%
Global Model (FedBoosting)	87%

VII. FUTURE SCOPE

Automating the process: Automation can be used in a federated learning system to control the selection and aggregation of models from various participants, as well as to keep track of the models' performance and modify the training process as necessary. Utilising the smart contract itself is one method for automating a blockchain-based federated learning system's workflow. Numerous steps in the federated learning process, including selecting and combining models, tracking performance, and allocating incentives, can

be automated using smart contracts.

Incentivization: Incentivization can be a crucial aspect of a blockchain-based federated learning system, as it encourages participants to contribute their data and models to the training process. There are several ways that incentivization can be included in a federated learning system, some of which are outlined below:

- **Reward-based system:** One approach to incentivization is to offer rewards to participants who contribute their data and models to the training process.
- **Reputation-based system:** Another approach to incentivization is to use a reputation-based system, where participants are rewarded based on their reputation within the community. Participants who contribute high-quality data and models are given a higher reputation score, which in turn entitles them to a greater share of the rewards.
- **Penalty-based system:** A penalty-based system can be used to incentivize participants to contribute high-quality data and models by penalizing them for low-quality contributions. Participants who contribute low-quality data or models may be subject to a penalty, such as a reduction in their reputation score or a reduction in their share of the rewards.

Storing the Model on the chain: Several benefits come with storing the model on the blockchain, including improved security, transparency, and immutability. Smart contracts can be used to store and manage machine learning models on the Internet Computer blockchain. The model can be stored in the smart contract, which can also be used to control the model training and updating procedure.

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